

Metal Contamination and Ecological Risks in the Kasai and Tshikapa Rivers (DR Congo): Insights from Geochemical Analyses of Sediments and Water

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Abstract: This study assessed the distribution of trace metals, rare earth elements (REEs), thorium (Th), and uranium (U) in sediments and surface waters of the Tshikapa and Kasai Rivers in Tshikapa, Democratic Republic of the Congo, and evaluated the associated contamination and potential ecological risks. Sediment physicochemical properties, grain-size distribution, trace elements, and REEs were determined using analytical techniques. Sediment pollution and contamination were assessed using the geoaccumulation index (Igeo), enrichment factor (EF), contamination factor (CF), degree of contamination (CD), ecological risk factor (Eri), and potential ecological risk index (RI), with reference to international sediment quality guidelines. Sediment metal concentrations generally followed the order $RT > RKAV > RKAM$. Cr, Cu, Zn, Cd, Pb, and Hg exceeded the corresponding Canadian Council of Ministers of the Environment (CCME) Interim Sediment Quality Guidelines at the investigated stations, while several concentrations also exceeded the probable effect levels, particularly at RT. In surface waters, Cd reached $4.2 \mu\text{g L}^{-1}$ at station RTL, exceeding the World Health Organization guideline value for drinking water. Total REE concentrations in sediments ranged from 111.1 to 298.4 mg kg^{-1} , with the highest values at RT, whereas dissolved REEs, Th, and U generally remained low in surface waters. Sediment Th and U concentrations ranged from 8.3–19.0 and 0.7–5.5 mg kg^{-1} , respectively. The degree of contamination exceeded 32 at all stations, while RI values ranged from 630.4 to 2335.3, indicating high to very high potential ecological risk, with the highest risk at RT and RKAV. Correlation patterns suggested that sediment organic matter and water content may influence the retention and distribution of several metals. The findings indicate significant metal contamination of river sediments and potential ecological concern, likely resulting from a combination of geogenic and anthropogenic inputs. Continued monitoring, source-apportionment studies, and appropriate measures to reduce contaminant inputs are recommended.

Keywords: Tshikapa River; Kasai River; Trace Metals; Sediment Contamination; Ecological Risk.

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I. INTRODUCTION

The degradation of aquatic ecosystems by inorganic micropollutants, particularly trace metal elements (TMEs), constitutes a major global environmental concern. This concern is exacerbated by their intrinsic toxicity, non-biodegradable nature, and propensity to bioaccumulate within food webs (Norvivor et al., 2024; Hao et al., 2022). Although natural mineral weathering processes contribute to the geochemical background levels (Yan et al., 2022; Sigue et al., 2025), anthropogenic inputs—including untreated industrial and mining effluents, hospital discharges, urban wastewater, and agricultural runoff—represent the predominant pathways of contamination (Hoang et al., 2021; Paithankar et al., 2021). In these riverine systems, sediments play a critical bimodal role: they initially act as sequestration sinks before potentially becoming secondary sources, thereby regulating the release and bioavailability of pollutants to benthic biota (Akonga et al., 2025; Cui et al., 2021; Hu et al., 2025; Jha et al., 2024).

Despite the existence of environmental legislative frameworks in developing countries, the enforcement of standards governing effluent discharges remains hindered by structural deficiencies in technical expertise and analytical monitoring infrastructure. This operational gap exposes local populations to chronic health risks associated with the ingestion of heavy metals (HMs) through the food chain (Yan et al., 2022; Sharma et al., 2024; Tarigan et al., 2025).

Although the scientific literature has documented various cases of contamination in Africa and the Democratic Republic of the Congo (DRC) (Kipelo et al., 2025), data on the Kasai River, a major tributary of the Congo River, remain fragmented, with the notable exception of the recent study by Manza et al. (2026). Nevertheless, the Kasai River is of vital socioeconomic and ecological importance, supporting

artisanal fishing, subsistence agriculture, and river transportation in southwestern DRC (Lesani et al., 2024).

The vulnerability of this ecosystem was tragically highlighted by the environmental crisis of July 2021. Massive mining discharges from Angola caused large-scale transboundary contamination, resulting in severe fish mortality and affecting nearly two million people. This pollution plume traveled several hundred kilometers downstream to Kwilu Province, directly threatening the food security of the 150,000 inhabitants of Bandundu City and the 7 million residents of the province, whose main source of animal protein depends on local fishery resources (CRREBaC, 2021).

In this context, the present study aims to (i) quantify the concentrations of trace elements (Hg, Pb, Cr, As, Cd, Cu, Ni, and Zn) in the sediment and water compartments of the Kasai and Tshikapa Rivers; and (ii) assess the degree of contamination by comparing these concentrations with international quality standards for the protection of the biological integrity of aquatic environments.

II. MATERIALS AND METHODS

➤ Description of the Study Area

The present study was conducted in the city of Tshikapa (Kasai Province, Democratic Republic of the Congo), located along the Kasai and Tshikapa Rivers (6°25'48" S; 20°47'12" E). The choice of these two watercourses was related to the transboundary pollution event that occurred in July 2021, following the contamination of a tributary of the Kasai River originating in Angola (CRREBaC, 2021) (Fig. 1). This event raised major environmental and public health concerns due to the potential risks to aquatic ecosystems and riparian populations that depend on these resources for domestic, fishing, and commercial activities.

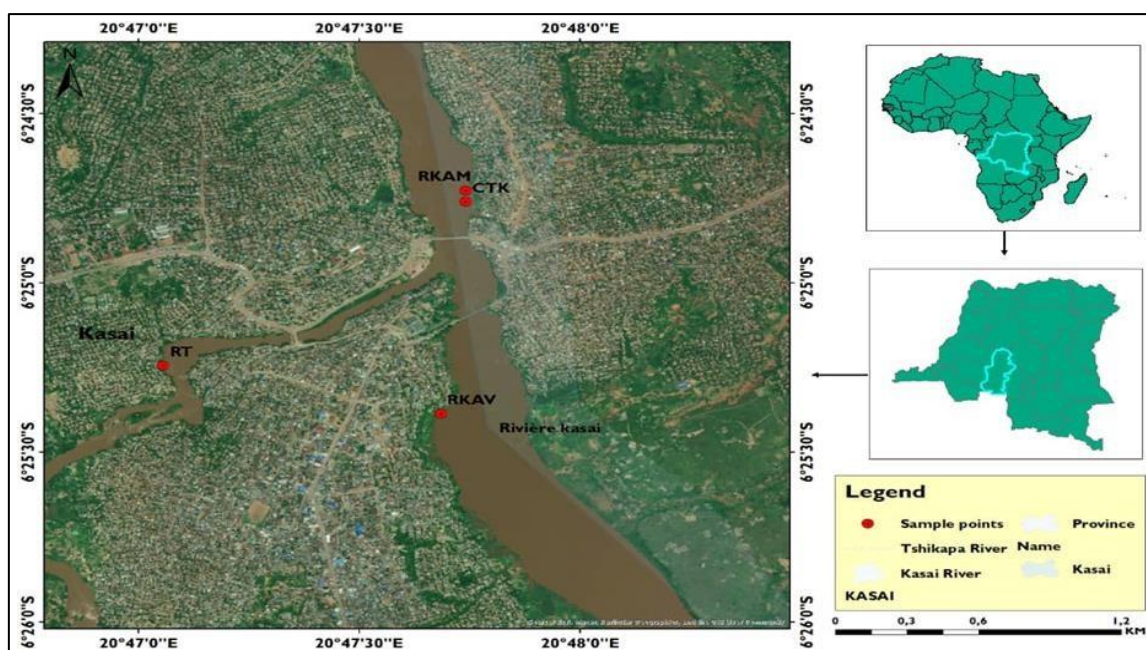


Fig 1 Location of the DR Congo in Africa; Location of Kasai Province within the DR Congo; Study Area with Sediment and Water Sampling Sites Along the Kasai and Tshikapa Rivers.

➤ Sampling Strategy and Sample Collection

Sampling campaigns were conducted between March and April 2025. Sampling involved surface sediments and river water collected from three stations distributed along the Tshikapa and Kasai Rivers, in the Kele, Sapide, and Abattoir neighborhoods. One station was located on the Tshikapa River and two on the Kasai River.

The sampling stations were coded as follows: RTs for the station located on the Tshikapa River, and RKAVs and RKAMs for the stations located on the Kasai River (Fig. 2). Composite sediment samples were coded RT, RKA, and RKAM, respectively, whereas water samples were designated RTL, RKAVL, RKAML, and CTK.



Fig 2 Photographic Representations of the Selected Sampling Stations : (A) Kasai River (RTs Station) ; (B) Tshikapa River (RKAVs and RKAMs Stations).

• Sediment Sampling

Surface sediment samples were collected manually at a depth ranging from 0 to 8 cm, approximately 5 m from the riverbank and at a water depth of less than 2 m. Each individual subsample consisted of approximately 50 g of sediment.

To ensure adequate spatial representativeness of the investigated stations, five subsamples were collected at each site using a centered-square sampling design, with an average spacing of approximately 10 m between sampling points. The subsamples were subsequently homogenized to obtain one representative composite sample for each station.

• Water Sampling

Water sampling, preservation, and subsequent analyses were performed in accordance with the standard methods of the American Public Health Association (APHA, 2022). At each station, water samples were collected in 500-mL glass

bottles that had previously been washed with detergent, soaked in a 10% nitric acid solution for 24 h, rinsed with double-distilled water, and dried at a temperature ranging from 90 to 95 °C (Boadi et al., 2020 ; APHA, 2022).

Prior to sampling, the bottles were rinsed with water from the corresponding site to minimize the risk of cross-contamination. Surface water samples were collected from the pelagic zone of each station, stored in a cooler containing ice, and immediately transported to the laboratory, where they were stored at 4 °C until analysis.

Samples intended for dissolved metal analysis were filtered in situ through a 0.45- μ m membrane, whereas samples intended for the determination of total recoverable metals were not filtered. All samples were subsequently acidified with nitric acid (HNO₃) to achieve a pH below 2, in accordance with the recommended analytical protocols.

➤ *Physicochemical Characteristics of Sediments*

After sediment homogenization, representative aliquots were isolated for physical and chemical analyses. Particle-size distribution, including clay, silt, and sand fractions, was determined by laser granulometry using a Coulter® LS-100 diffractometer (Beckman Coulter, Fullerton, CA, USA), following the protocol established by Mabidi et al. (2024). According to the nomenclature of Blott and Pye (2001), the textural classification of the samples was based on mean particle diameter: particles smaller than 2 µm were classified as clay, whereas the silt fraction ranged from 2 to 63 µm. Particles with diameters ranging from 63 µm to 2 mm were classified as sand.

Water content (WC) was quantified gravimetrically by measuring the mass loss of aliquots of fresh sediment after drying at 105 °C for approximately 24 h, until constant mass was reached. In parallel, the organic matter (OM) fraction was determined using the loss-on-ignition (LOI) method. Aliquots of previously dehydrated sediment were subjected to combustion in a muffle furnace (Salvis AG, Luzern, Switzerland) at 550 °C for 1 h (Poté et al., 2008; Deko et al., 2026). This approach provides an estimate of the total organic matter content through the thermal oxidation of carbonaceous compounds.

➤ *Pretreatment of Sediment and Water Samples*

Sediment samples were freeze-dried, ground, homogenized, and then sieved through a 63-µm mesh. Acid digestion was performed according to the protocols described by Thevenon et al. (2013).

Approximately 10–15 mg of sediment powder was completely digested in Teflon vessels placed on a glass-ceramic hotplate using ultrapure acids. The procedure consisted of three successive steps: (i) addition of 1 mL of HNO₃ (Suprapur®, 65%); (ii) addition of a mixture consisting of 0.5 mL of HClO₄ (Suprapur®, 70%) and 0.5 mL of HF (Suprapur®, 40%); and (iii) final treatment with 0.5 mL of HNO₃ (Suprapur®, 65%).

Between each step, the solvents were evaporated to near dryness. The resulting residues were then redissolved in 10 mL of a 1% HNO₃ solution prior to chemical analysis. Analyses were performed within 24 h after dilution.

For water samples, 10-mL aliquots of acidified water were filtered using 0.45-µm membranes (Millex®-LCR, Millipore). Appropriate dilutions were subsequently prepared using a 1% HNO₃ solution prior to instrumental analysis.

➤ *Analysis of Trace Metals and Rare Earth Elements in Sediments and Water*

The concentrations of trace metals and rare earth elements (REEs) in sediments and water were determined by inductively coupled plasma mass spectrometry (ICP-MS; Agilent 7700x series), which is suitable for the analysis of complex environmental matrices.

To minimize spectral interferences that could affect the quality of the analytical results, a helium-mode

collision/reaction cell and interference-correction equations were used. Instrument calibration was performed using multielement standard solutions prepared at different concentrations (0, 0.02, 1, 5, 20, 100, and 200 µg L⁻¹), according to the protocols described by Poté et al. (2008) and Thevenon et al. (2013).

Analytical quality and measurement reliability were verified using the certified reference materials TMDA-70 and LKSD4 for water and sediment analyses, respectively. Relative standard deviations obtained from three replicate measurements remained below 10%, whereas analytical blanks accounted for less than 2% of the sample signal.

The concentrations measured in water and sediment samples are expressed in µg L⁻¹ and mg kg⁻¹ dry weight, respectively.

➤ *Analysis of Total Mercury in Sediments*

Total mercury (Hg) concentrations in sediments were determined using a direct mercury analyzer (DMA-80 III, MWS GmbH) operating according to the principle of cold vapor atomic absorption spectrometry (CV-AAS).

The analytical procedure involved sample combustion, amalgamation of mercury onto a gold trap, and subsequent quantification of gaseous mercury by atomic absorption. The method recovery rate was greater than 99%, with a relative uncertainty of approximately 4%. The detection limit, calculated as the mean blank value plus three standard deviations, was 0.003 µg g⁻¹. Analytical reproducibility was high, with variations among replicates remaining below 5%.

Analytical quality control was ensured using the certified reference material SRM 2702 (National Institute of Standards and Technology, USA), which was analyzed simultaneously with the samples according to the Feststoffe Schnell protocol (without a drying step; combustion temperature set at 750 °C). The results showed an Hg recovery rate of 99% relative to the certified value. Total mercury concentrations are expressed as mg kg⁻¹ dry weight of sediment.

➤ *Statistical Analyses*

All analyses were performed in triplicate. Statistical data processing, including Spearman correlation analyses, was performed using XLSTAT software (Addinsoft, 2020, New York, USA).

➤ *Assessment of the Nature and Degree of Sediment Pollution*

The nature and degree of sediment contamination were assessed using the enrichment factor (EF) and geoaccumulation index (I_{geo}), following the approach described by Maanan et al. (2004).

• *Enrichment Factor (EF)*

The enrichment factor was calculated to discriminate the anthropogenic origin of trace metals using the following equation: $EF = (Metal/Sc)_{Sample} / (Metal/Sc)_{Background}$.

Where “Metal” represents the concentration of the metal of interest in the sample or in the geochemical background, and “Sc” represents the concentration of scandium used as the normalizing element. Upper Continental Crust (UCC) values proposed by McLennan (2001) were used as geochemical background values.

- *Geoaccumulation Index (IGEO)*

The degree of sediment pollution was assessed using the geoaccumulation index according to the following equation:

$$I_{geo} = \log_2 [C_n / 1.5(B_n)]$$

Where “C_n” corresponds to the concentration of metal n in the sediment and “B_n” to its geochemical background concentration. The factor 1.5 was introduced to account for natural variations in the lithogenic background (Akonga et al., 2026).

➤ *Ecological Risk Assessment*

The ecological risks associated with trace metals in sediments were assessed using four indicators: the contamination factor (CF), degree of contamination (CD), individual ecological risk index (Eri), and potential ecological risk index (RI) (Deko et al. 2026).

- *Contamination Factor (CF)*

The contamination factor was calculated using the following equation:

$$CF = C_n / B_n$$

Where “C_n” represents the concentration of the metal in the sample and “B_n” its concentration in the geochemical background (Rubio et al., 2000; Förstner et al., 1989).

- *Degree of Contamination (CD)*

The degree of polymetallic contamination was determined from the sum of the contamination factors:

$$CD = \sum CF_i$$

Where “CF_i” represents the contamination factor of metal i. Eight trace metals were considered in the calculation of CD.

- *Individual Ecological Risk Index (Eri)*

The individual ecological risk index was calculated using the following equation:

$$Eri = Tri \times CF_i$$

Where “Tri” represents the toxic-response factor of the metal under consideration and “CF_i” its contamination factor. The Tri values used were 40, 30, 10, 5, 5, 5, 2, and 1 for Hg, Cd, As, Co, Cu, Pb, Cr, and Zn, respectively (Håkanson, 1980; Islam et al., 2014).

- *Potential Ecological Risk Index (RI)*

The potential ecological risk index was determined by summing the individual ecological risk indices:

$$RI = \sum Eri$$

Where “Eri” corresponds to the ecological risk index associated with each metal. This index simultaneously considers the level of contamination, metal toxicity, and the ecological sensitivity of exposed organisms (Singh et al., 2002).

Finally, the ecotoxicological significance of the metal concentrations measured in sediments was assessed by comparing them with the threshold values of the Sediment Quality Guidelines (SQGs) and Probable Effect Levels (PELs) established by the Canadian Council of Ministers of the Environment (CCME, 1999). These guidelines allow the identification of concentrations that may cause adverse biological effects on aquatic organisms.

III. RESULTS AND DISCUSSION

➤ *Sediment Organic Matter and Water Content*

The main physicochemical characteristics of the sediment samples, including water content (WC), total organic matter (OM), and particle-size distribution, are presented in Table 1.

Table 1 Water Content, Organic Matter, and Particle-Size Distribution of Sediments

Sample	WC	OM	Clay	Silt	Sand
RT	24.8	7.72	1.42	58.59	39.99
RKAV	20.95	5.31	1.18	22.77	76.05
RKAM	21.84	2.2	0.78	13.9	85.32

Sediment water content ranged from 20.95% at RKAV to 24.80% at RT, whereas TOM contents ranged from 2.20% at RKAM to 7.72% at RT. Compared with uncontaminated freshwater sediments, for which organic matter contents are generally reported to range from 0.1 to 6% (Poté et al., 2008), the values recorded at RKAV and RKAM were relatively low and consistent with weakly impacted conditions. In contrast, the value at RT was slightly above this range, suggesting moderate organic matter enrichment. Nevertheless, the observed values remain within the range reported by Manza

et al. (2026) for sediments from the Kasai River (1.26–12.42%), highlighting the variability of organic carbon pools in tropical riverine environments.

Particle-size analysis showed a predominance of sand and silt fractions, whereas the clay fraction remained low in all samples (<1.5%). Sand contents ranged from 39.99% at RT to 85.32% at RKAM, while silt contents ranged from 13.90% at RKAM to 58.59% at RT. According to the USDA classification (Groenendyk et al., 2015), the sediments from

the Kasai River correspond to fine silt, whereas those from the Tshikapa River are classified as silty sand.

Although quantitatively less abundant, fine particles, particularly silt and clay, may play an important role in trace-metal retention because of their relatively high specific surface area and cation-exchange capacity (Zhu et al., 2024; Bonciani et al., 2025). Accordingly, the relatively high silt and organic matter contents at RT may provide favorable conditions for the adsorption and accumulation of metal

contaminants. Conversely, the predominance of sandy particles at RKAV and RKAM may be associated with a comparatively lower capacity for contaminant retention.

➤ Trace-Metal Concentrations in Sediments and Water

The concentrations of selected trace elements (Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, As, Se, Ag, Cd, Th, Pb, and Hg) measured in sediments and surface waters from the Tshikapa and Kasai Rivers are presented in Table 2.

Table 2 Concentration of Heavy Metals in Sediments (mg kg^{-1} Dry Weight) and River Water ($\mu\text{g L}^{-1}$) from the Kasai and Tshikapa Rivers.

Sediment samples (mg kg^{-1})																
	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	As	Se	Ag	Cd	Th	Pb	Hg
RT	1095.6	83	187.9	666.5	52480.5	40.4	293.6	322.4	473.1	1.7	5.3	0.1	2.2	176.7	129.1	2.1
RKAV	394.3	50.6	74.3	501.7	30787.7	24.4	31.9	187.1	235.3	0.8	1.8	0	1.8	52.9	61	1.4
RKAM	263	34.6	46.8	394.1	24894.8	18.2	15.4	164.6	286.8	0.9	2	0	1.1	39.5	65.8	0.3
SQGs			37.3					35.7	123	5.9			0.6		35	0.17
PELs			90					197	315	17			3.5		91.3	0.486
Water samples ($\mu\text{g L}^{-1}$)																
	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	As	Cd	Ba			Pb	
RTL	3.9	0.5	0.3	1.1	204.4	0.1	0.4	50.3	22.8	0.1	4.2	27.0			6.6	
RKAVL	0.9	0.2	0.1	0.6	47.3	0.1	1.0	20.8	37.4	0.1	1.3	22.3			<0	
RKAML	1.6	0.2	0.1	<0	32.2	0.1	0.3	22.8	23.3	0.1	1.2	14.5			4.6	
CTK	7.8	0.8	0.5	1.4	257.2	0.1	0.4	26.8	13.4	0.1	2.7	19.0			3.7	
TMDA	40.4	58.6	67.0	81.7	83.4	67.2	45.7	68.5	86.6	34.4	72.5	39.3			58.8	
REFERENC E*			50	400			70	2000		10	3				10	

Sediment concentrations were compared with the Canadian Council of Ministers of the Environment (CCME) Interim Sediment Quality Guidelines (ISQGs) and Probable Effect Levels (PELs) for the protection of freshwater aquatic life. The CCME values used in this study are expressed on a dry-weight basis. For example, the freshwater ISQG and PEL values are 37.3 and 90 mg kg^{-1} for Cr, 35.7 and 197 mg kg^{-1} for Cu, 123 and 315 mg kg^{-1} for Zn, and 0.6 and 3.5 mg kg^{-1} for Cd.

Several trace metals, particularly Cr, Cu, Zn, Cd, Pb, and Hg, exceeded their corresponding CCME ISQG values in the sediment samples. Cr concentrations ranged from 46.8 mg kg^{-1} at RKAM to 187.9 mg kg^{-1} at RT, exceeding the freshwater ISQG of 37.3 mg kg^{-1} . Similar exceedances were observed for Cu (164.6–322.4 mg kg^{-1} ; ISQG = 35.7 mg kg^{-1}), Zn (235.3–473.1 mg kg^{-1} ; ISQG = 123 mg kg^{-1}), Cd

(1.1–2.2 mg kg^{-1} ; ISQG = 0.6 mg kg^{-1}), Pb (61.0–129.1 mg kg^{-1} ; ISQG = 35 mg kg^{-1}), and Hg (0.3–2.1 mg kg^{-1} ; ISQG = 0.17 mg kg^{-1}). Fe concentrations were also relatively high (24,894.8–52,480.5 mg kg^{-1}), consistent with the ferrallitic and ferruginous geological context of the Kasai Basin.

Some concentrations also exceeded the corresponding PELs, indicating a greater likelihood of adverse biological effects according to the CCME framework. The highest values were recorded at RT for Cr (187.9 mg kg^{-1}), Cu (322.4 mg kg^{-1}), Zn (473.1 mg kg^{-1}), and Pb (129.1 mg kg^{-1}), while Hg exceeded the PEL at RKAV (1.4 mg kg^{-1}) and RT (2.1 mg kg^{-1}). These exceedances indicate that benthic organisms may be exposed to potentially adverse levels of several contaminants, particularly at RT.

Sedimentary metal concentrations followed the decreasing order $RT > RKAV > RKAM$, suggesting a greater degree of metal accumulation at RT. This spatial pattern may reflect the combined influence of natural geochemical characteristics and anthropogenic inputs. Sediments act as long-term repositories for contaminants and can therefore provide an integrated indication of metal accumulation over time (Thevenon et al., 2011). The observed spatial variability is also consistent with differences in sediment physicochemical properties, particularly organic matter content and particle-size distribution. Organic matter can enhance metal retention through adsorption and complexation involving humic substances (Jokinen et al., 2020; Yan et al., 2022), while fine particles provide abundant reactive surfaces for contaminant sorption. Similar mechanisms have been reported in other aquatic systems, where organic matter and mineral phases contribute to metal stabilization through adsorption, complexation, and precipitation processes (Bastianini et al., 2024).

The enrichment observed at RT, particularly for Cr, Cu, Zn, Pb, and Hg, may therefore reflect a combination of sediment geochemical properties and local anthropogenic inputs. However, the available data do not allow a definitive separation of geogenic and anthropogenic sources. The relatively low concentrations of some elements at individual stations may reflect differences in lithology, hydrodynamics, particle sorting, and contaminant inputs. Additional source-apportionment approaches would be required to identify the relative contribution of individual anthropogenic sources.

In surface waters, trace-metal concentrations showed marked spatial variability, following the order $CTK > RTL > RKAVL > RKAML$ (Table 2).

Fe was the dominant element, with concentrations ranging from $32.2 \mu\text{g L}^{-1}$ at RKAML to $257.2 \mu\text{g L}^{-1}$ at CTK. This predominance is consistent with the ferrallitic and ferruginous geological context of the Kasai Basin. Cu concentrations ranged from 20.8 to $50.3 \mu\text{g L}^{-1}$, Zn from 13.4 to $37.4 \mu\text{g L}^{-1}$, and Ba from 14.5 to $27.0 \mu\text{g L}^{-1}$.

Cr (0.1 – $0.5 \mu\text{g L}^{-1}$), Ni (0.3 – $1.0 \mu\text{g L}^{-1}$), and Pb (3.7 – $6.6 \mu\text{g L}^{-1}$) occurred at relatively low concentrations. In contrast, Cd represented the most concerning element, with concentrations ranging from $1.2 \mu\text{g L}^{-1}$ at RKAML to $4.2 \mu\text{g L}^{-1}$ at RTL. The Cd concentration at RTL exceeded the WHO guideline value reported for drinking water, indicating that untreated river water from this location may not be suitable for direct human consumption. The relatively high concentrations of Fe, Cu, Cd, and Pb at RTL may reflect local inputs associated with domestic, artisanal, or mining activities and/or remobilization from contaminated sediments.

Comparable trends have been reported in other tropical river systems. Norvivor et al. (2024), for example, reported a predominance of Fe among dissolved metals in the Nubui River. Their Pb concentrations ranged from 0.001 to 0.022 mg L^{-1} , Cd from 0.020 to 0.035 mg L^{-1} , and Zn from 0.042 to 0.067 mg L^{-1} . The Pb concentrations measured in the Tshikapa and Kasai Rivers (0.0037 – 0.0066 mg L^{-1}) were lower than the upper values reported for the Nubui River, while Zn concentrations (0.0134 – 0.0374 mg L^{-1}) were also generally lower. Cd at RTL, however, reached a comparatively concerning level, emphasizing the importance of monitoring this element in anthropogenically influenced tropical river systems.

Overall, the predominance of Fe in the dissolved phase appears compatible with the regional geological and pedological setting, whereas the localized enrichment of Cd, Pb, and Zn may indicate additional anthropogenic contributions. Nevertheless, these interpretations should be considered indicative rather than definitive because the available dataset does not allow rigorous source apportionment.

➤ Rare Earth Element Concentrations in Sediments and Water

The concentrations of the fourteen rare earth elements (REEs), namely La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, and Lu, together with Th and U, are presented in Table 3.

Table 3 Concentration of Rare Earth Elements in Sediments (mg kg^{-1} dry Weight) and River Water ($\mu\text{g L}^{-1}$) from Kasai and Tshikapa Rivers.

Sample	Sediment samples (mg kg^{-1})													Th	U
	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Ho	Tm	Yb	Lu	ΣREE		
RT	56.2	153.2	13.6	51.4	8.7	1.9	8.0	1.0	1.1	0.4	2.5	0.4	298.4	19.0	5.5
RKAV	20.7	54.4	4.7	17.3	2.9	0.6	2.6	0.3	0.3	0.1	0.8	0.1	104.8	6.4	0.7
RKAM	19.0	60.3	4.6	17.5	3.3	0.6	3.0	0.4	0.5	0.2	1.5	0.2	111.1	8.3	1.3
STSD 4	30.0	57.1	8.5	35.0	7.1	2.1	7.0	1.0	1.1	0.4	2.8	0.4		4.5	3.9
NASC*	32	73	7.9	33	5.7	1.24	5.2	0.85	1.04	0.50	3.1	0.48		-	-
	Water samples ($\mu\text{g L}^{-1}$)													Th	U
	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Ho	Tm	Yb	Lu			
RT	0.1	0.3	0	0.1	0	0	0	0	0	0	0	0	0	0	0.1
RKAV	0.2	0.1	0	0.1	0	0	0	0	0	0	0	0	0	0	0.1
RKAM	0.1	0.1	0	0.1	0	0	0	0	0	0	0	0	0	0	0
CTK	0.2	0.4	0.1	0.2	0	0	0	0	0	0	0	0	0	0	0.1
TMDA	0	0	0	0	0	0	0	0	0	0	0	0	0	149.3	

In sediments, individual REE concentrations ranged from 0.4 to 153.2 mg kg⁻¹ at RT, 0.1 to 54.4 mg kg⁻¹ at RKAV, and 0.2 to 60.3 mg kg⁻¹ at RKAM. Overall, higher concentrations were observed at RT. The ΣREE concentrations ranged from 111.1 to 298.4 mg kg⁻¹, with the highest value recorded at RT. The value at RT exceeded the upper continental crust reference reported by McLennan (2001), suggesting relative REE enrichment at this station.

The observed variability may reflect differences in lithological inputs, mineralogical composition, sediment sorting, and grain size. In particular, the predominance of sandy, silica-rich fractions at some locations may result in dilution of REE concentrations relative to finer, more reactive sediment fractions. Such effects have been reported in other sedimentary systems (Sager et al., 2024).

Sedimentary Th and U concentrations ranged from 8.3 to 19.0 mg kg⁻¹ and from 0.7 to 5.5 mg kg⁻¹, respectively. These values exceeded the reported global soil averages of approximately 5 mg kg⁻¹ for Th and 1 mg kg⁻¹ for U (Zlobina et al., 2022), suggesting relative enrichment of these naturally occurring radionuclides.

In surface waters, REEs, Th, and U occurred at very low concentrations, generally below 0.5 µg L⁻¹ (Table 3). The dissolved REE pattern was dominated by LREEs, particularly La, Ce, and Nd, whereas most HREEs were below the analytical detection limits. Such a pattern is consistent with

the strong association of REEs with dissolved organic matter, colloids, and particulate phases in natural surface waters.

The low dissolved concentrations contrast strongly with the relatively high concentrations measured in sediments, indicating limited dissolved mobility. Th and U were also detected only at trace levels, generally ≤0.1 µg L⁻¹. In particular, dissolved U concentrations were far below the WHO guideline value of 30 µg L⁻¹ for drinking water. These results suggest that most of the REEs and naturally occurring radionuclides are preferentially associated with the particulate/sedimentary phase rather than remaining in solution.

The contrast between sedimentary enrichment and low dissolved concentrations suggests strong partitioning of these elements between solid and aqueous phases. Adsorption onto Fe and Mn oxyhydroxides, association with particulate organic matter, and incorporation into mineral phases may contribute to their retention in sediments. Consequently, suspended particles and sediment transport may constitute important pathways for downstream transfer of these elements in the river system.

➤ Assessment of Metal Pollution and Ecological Risk

The geoaccumulation index (Igeo) and enrichment factor (EF) were used to assess the degree of sediment contamination and to provide an indication of possible anthropogenic contributions (Akonga et al., 2025). The results are presented in Table 4.

Table 4 Calculated IGEO and EF Pollution Indices: Results for Trace Metals in Sediment Samples

Igeo	RT	RKAV	RKAM	
Ti	11.2	9.8	9.2	
V	-0.1	-0.8	-1.4	
Cr	1.8	0.5	-0.2	
Mn	-0.4	-0.8	-1.2	
Fe	0.0	-0.8	-1.1	
Co	1.4	0.7	0.3	
Ni	3.3	0.1	-1.0	
Cu	3.4	2.6	2.5	
Zn	2.2	1.1	1.4	
As	-0.4	-1.5	-1.3	
Se	-3.8	-5.4	-5.2	
Ag	0.4	-	-	
Cd	3.9	3.6	2.9	
Th	-2.2	-1.3	-0.9	
Pb	2.1	1.0	1.1	
Hg	4.6	4.1	1.8	

EF	RT	RKAV	RKAM	
Ti	2058.8	2212.3	1125.0	
V	0.8	1.4	0.7	
Cr	3.0	3.6	1.7	
Mn	0.6	1.4	0.8	
Fe	0.8	1.5	0.9	
Co	2.3	4.1	2.3	
Ni	8.3	2.7	1.0	
Cu	9.1	15.7	10.6	
Zn	3.8	5.6	5.2	

As	0.6	0.9	0.8	EF 25-50 – very severe enrichment EF > 50 – extremely severe enrichment
Se	0.1	0.1	0.1	
Ag	1.1	0.0	0.0	
Cd	12.7	30.9	14.4	
Th	0.1	0.1	0.1	
Pb	3.6	5.1	4.2	
Hg	21.1	42.1	6.9	

The Igeo values showed substantial spatial variability. At RT, V, Mn, Fe, As, Se, Ag, and Th generally fell within the practically unpolluted to moderately polluted categories, whereas Cr and Co indicated moderate pollution. Zn and Pb indicated moderate to strong pollution, while Ni, Cu, and Cd were strongly polluted. Hg reached the strongly to extremely polluted category, whereas Ti exhibited an exceptionally high Igeo value corresponding to extreme enrichment.

At RKAV, most elements showed low to moderate contamination, while Zn and Pb indicated moderate pollution, Cu moderate to strong pollution, Cd strong pollution, Hg strong to extreme pollution, and Ti extreme pollution. RKAM displayed relatively lower Igeo values for most elements, although Cu and Cd remained moderately to strongly polluted and Ti remained extremely enriched.

The EF values provided further evidence of differential enrichment among the elements. Several elements, including V, Mn, Fe, As, Se, Ag, and Th, showed little or no enrichment, whereas Cr, Ni, Cu, Zn, Cd, Pb, and Hg

displayed moderate to very high enrichment at one or more stations.

However, the extremely high EF values obtained for Ti require cautious interpretation. Ti is commonly associated with resistant mineral phases and may be strongly influenced by local lithology and sediment mineralogy. Therefore, the Ti enrichment observed in this study should not be interpreted exclusively as evidence of anthropogenic contamination without additional mineralogical or source-apportionment evidence.

For Cu, Zn, Cd, Pb, and Hg, the combination of elevated concentrations, Igeo values, and EF values is consistent with an additional anthropogenic contribution. Nevertheless, the possibility of geogenic control cannot be completely excluded, particularly in the absence of a detailed local geochemical background.

The contamination factor (CF), degree of contamination (CD), individual ecological risk factor (Eri), and potential ecological risk index (RI) are presented in Table 5.

Table 5 Results for the Ecological Risk Indices CF, CD, Eri, and RI Calculated for Trace Elements in the Sediment Samples

Samples	CF																CD
	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	As	Se	Ag	Cd	Th	Pb	Hg	
RT	3652	1.4	5.4	1.1	1.5	4	14.7	16.1	6.7	1.1	0.1	2	22.4	0.3	6.5	37.5	93.3
RKAV	1314.3	0.8	2.1	0.8	0.9	2.4	1.6	9.4	3.3	0.5	0	0	18.4	0.6	3.1	25	64.2
RKAM	876.7	0.6	1.3	0.7	0.7	1.8	0.8	8.2	4	0.6	0	0	11.2	0.8	3.3	5.4	35.9

CD classification	
$8 \leq DC < 16$ – moderate contamination	
$16 \leq DC < 32$ – high contamination	
$32 \leq DC$ – very high contamination	
CF classification	
$FC < 1$ – low contamination	
$1 < FC < 3$ – moderate contamination	
$3 < FC < 6$ – considerable contamination	
$6 < FC$ – very high contamination	

Eri									RI
Samples	Hg	Cd	As	Co	Cu	Pb	Cr	Zn	
RT	1500	673.5	11.3	20.2	80.6	32.3	10.7	6.7	2335.3
RKAV	1000	551	5.3	12.2	46.8	15.3	4.2	3.3	1638.1
RKAM	214.3	336.7	6	9.1	41.2	16.5	2.7	4	630.4

Eri classification	
$Eri < 40$ – low ecological risk	
$40 < Eri < 80$ – moderate ecological risk	
$80 < Eri < 160$ – considerable ecological risk	
$160 < Eri < 320$ – high ecological risk	
$Eri > 320$ – very high ecological risk	

RI classification
IRI<150 – low ecological pollution level
150≤RI<300 – moderate ecological pollution level
300≤RI<600 – severe ecological pollution level
RI>600 – serious ecological pollution level

CF values indicated considerable to very high contamination for several elements at RT, particularly Ti, Cr, Co, Ni, Cu, Zn, Cd, Pb, and Hg. At RKAV, considerable to very high contamination was mainly associated with Ti, Cu, Zn, Cd, Pb, and Hg, whereas at RKAM the strongest contamination was mainly associated with Ti and Cd.

The CD values exceeded 32 at all three stations, indicating a very high overall degree of contamination according to the classification adopted in this study.

The individual ecological risk factors were generally low for As, Co, Cr, Pb, and Zn. Cu reached a considerable ecological risk level at RT, whereas Cd and particularly Hg contributed substantially to the ecological risk at some stations. The potential ecological risk index (RI) was 2335.3 at RT, 1638.1 at RKAV, and 630.4 at RKAM. Based on the

classification used in this study, these values correspond to very high potential ecological risk at RT and RKAV and high potential ecological risk at RKAM.

Overall, the ecological risk assessment indicates substantial alteration of sediment quality, particularly at RT and RKAV. The results highlight the need for further monitoring of potentially toxic metals, especially Cd, Hg, Cu, Zn, and Pb, in sediments and associated aquatic biota.

➤ Spearman Rank Correlation Analysis

Spearman's rank correlation analysis was performed to explore relationships between trace-metal concentrations and the main physicochemical characteristics of the sediments, particularly water content (WC) and organic matter (OM) (Table 6).

Table 6 Spearman's Rank Correlation Matrix Between Trace Metals and Physicochemical Parameters in Sediment Samples

Variab les	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	As	Se	Ag	Cd	Th	Pb	Hg
WC	0.50 0	0.50 0	0.50 0	0.50 0	0.50 0	0.50 0	0.50 0	0.50 0	1.00 0	1.00 0	1.00 0	0.86 6	0.50 0	0.50 0	1.00 0	0.50 0
OM	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Ti		1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
V			1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Cr				1.00 0	1.00 0	1.00 0	1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Mn					1.00 0	1.00 0	1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Fe						1.00 0	1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Co							1.00 0	1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Ni								1.00 0	0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Cu									0.50 0	0.50 0	0.50 0	0.86 6	1.00 0	1.00 0	0.50 0	1.00 0
Zn										1.00 0	1.00 0	0.86 6	0.50 0	0.50 0	1.00 0	0.50 0
As											1.00 0	0.86 6	0.50 0	0.50 0	1.00 0	0.50 0
Se												0.86 6	0.50 0	0.50 0	1.00 0	0.50 0
Ag													0.86 6	0.86 6	0.86 6	0.86 6
Cd														1.00 0	0.50 0	1.00 0
Th															0.50 0	1.00 0

Pb																	0.50 0
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The correlation matrix showed several strong positive associations among the analyzed variables. Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Th, and Hg displayed very high correlations with one another and with OM, whereas Zn, As, and Pb showed strong relationships with water content and with one another. Ag displayed an intermediate pattern, with relatively high correlations with several metals and with the sediment physicochemical parameters.

The observed association between several metals and OM suggests that organic matter may contribute to metal retention through adsorption and complexation processes. Similarly, the positive relationships between water content and several metals may reflect the influence of sediment moisture, pore-water conditions, and fine-particle accumulation on contaminant distribution.

However, these correlations must be interpreted with considerable caution because only three sediment samples were available for the correlation analysis. With such a very small sample size, Spearman coefficients are strongly constrained and may attain extreme values. Consequently, the observed correlations should be regarded as exploratory patterns of co-variation rather than statistically robust evidence of common contamination sources or causal geochemical relationships.

The strong associations among several metals may nevertheless indicate similar geochemical behavior or common controlling factors. These observations are broadly consistent with Manza et al. (2026), who reported strong associations between organic matter and several trace metals in sediments from the Kasai Basin, including Cd, Pb, Hg, As, and Co. Organic matter can promote metal retention through complexation with organic ligands and adsorption onto organic-rich particles.

Manza et al. (2026) also identified carbonate phases as potential secondary regulators of Cd, Zn, Pb, and Cu through adsorption and co-precipitation mechanisms. Although carbonate contents were not quantified in the present study, the observed relationships with OM suggest that organic matter may represent one of the important factors controlling metal retention in these sediments.

The correlation analysis suggests that sediment physicochemical properties, particularly organic matter and water content, may influence the spatial distribution and retention of trace metals. Nevertheless, additional samples are required to statistically validate these relationships and to distinguish between common sources, shared geochemical controls, and coincidental correlations.

IV. CONCLUSION

This study showed significant spatial variability in the physicochemical characteristics and contamination levels of sediments and surface waters from the Kasai and Tshikapa

Rivers. Sediments were mainly composed of sand and silt, with clay contents below 1.5% and organic matter ranging from 2.20 to 7.72%. Several potentially toxic metals exceeded the CCME sediment quality guidelines, particularly Cr (46.8–187.9 mg kg⁻¹), Cu (164.6–322.4 mg kg⁻¹), Zn (235.3–473.1 mg kg⁻¹), Cd (1.1–2.2 mg kg⁻¹), Pb (61.0–129.1 mg kg⁻¹), and Hg (0.3–2.1 mg kg⁻¹). The highest concentrations were generally recorded at RT, indicating greater metal accumulation at this station.

The pollution indices confirmed substantial sediment contamination, particularly by Cu, Zn, Cd, Pb, and Hg. The potential ecological risk index (RI) reached 2335.3 at RT, 1638.1 at RKAV, and 630.4 at RKAM, corresponding to very high risk at RT and RKAV and high risk at RKAM. REEs were also relatively enriched in sediments, with ΣREE concentrations ranging from 111.1 to 298.4 mg kg⁻¹ and the highest value at RT. In contrast, dissolved REEs, Th, and U were generally below 0.5 µg L⁻¹, indicating strong partitioning toward the sedimentary and particulate phases. In surface waters, Cd was of particular concern, reaching 4.2 µg L⁻¹ at RTL and exceeding the WHO guideline for drinking water.

The results indicate that the sediments of the studied river system constitute important reservoirs of potentially toxic metals and REEs, with RT being the most affected station. The observed enrichment of Cu, Zn, Cd, Pb, and Hg may reflect a combination of natural geochemical characteristics and anthropogenic inputs, although definitive source attribution requires further investigation. Continued monitoring of sediments, water, and aquatic organisms is therefore recommended, together with additional sampling, mineralogical characterization, local background determination, and source-apportionment studies to better distinguish geogenic from anthropogenic contributions and assess the associated environmental and human-health risks.

DECLARATIONS

➤ Ethics Approval

- Not applicable.

➤ Consent to Participate

- Not applicable.

➤ Consent to Publish

- Not applicable.

REFERENCES

- [1]. Akonga, M-M.O., Gizanga, R.V., Musibono, D.E., Carvalho, F.P., Poté, J., Atibu, E.K., 2026. Accumulation of toxic metals in sediments and oligochaetes around urban uncontrolled landfills in

- Kinshasa, DR Congo. *Environ Geochem Health* 48:120 <https://doi.org/10.1007/s10653-026-02980-9>
- [2]. Akonga, M.-M., Mutwejile, H. M., Gizanga, R. V., Kayembe, J. S., Musibono, D. E., Carvalho, F. P., Poté, J., & Atibu, E. K., 2025. Bioaccumulation of metals and metalloids in soils and in the earthworm *Hyperodrilus africanus* from uncontrolled landfills in Kinshasa, Democratic Republic of the Congo. *Journal of Toxicological Studies*, 3(1), Article 2115. <https://doi.org/10.59400/jts2115>
- [3]. APHA (American Public Health Association), 2022. Standard Methods for the Examination of Water and Wastewater. 24th Edition, American Public Health Association. <https://www.scirp.org/reference/referencespapers?referenceid=3901712>
- [4]. Bastianini, L., Rogerson, M., Brasier, A., Prior, T. J., Hardman, K., Dempsey, E., Bird, A., & Mayes, W. M., 2024. Ikaite formation in streams affected by steel waste leachate: First report and potential impact on contaminant dynamics. *Chemical Geology*. <https://doi.org/10.1016/j.chemgeo.2023.121842>
- [5]. Blott, S.J., Pye, K., 2001. Gradistat: A grain size distribution and statistics package for the analysis of unconsolidated sediments. *Earth Surf Process Landforms*. 26:1237–48. <https://doi.org/10.1002/esp.261>.
- [6]. Boadi, N. O., Saah, S. A., Baa-Poku, F., Mensah, E. A., and Addo, M., 2020. Safety of borehole water as an alternative drinking water source. *Scientific African*, 10:e00657. doi: 10.1016/j.sciaf.2020.e00657
- [7]. Bonciani, N., Lauridsen, B.W., Jakobsen, R., Feilberg, K.L., 2025. Impact of offshore energy activities on trace elements content and mobility in marine sediments. *Mar Pollut Bull.* 2025. <https://doi.org/10.1016/j.marpolbul.2024.117491>.
- [8]. CCME, 1999. Canadian sediment quality guidelines for the protection of aquatic life. EPC 98-E, Canadian environmental quality guidelines. Canadian Council of Ministers of the Environment. Available online: https://supp0rt.esdat.net/Environmental%20Standards/canada/ccme/sedqg_summary_table.pdf. Accessed 3 Jun 2025.
- [9]. CRREBaC, 2021. Water pollution disaster in Tshikapa and Kasai rivers identified in the CB-CIS tool: CRREBaC's call for action. Ref. No.: CRREBaC/RTM/043/082021, Kinshasa, 13 August 2021. Available online: <https://medd.gouv.cd/wp-content/uploads/2021/08/Pollution-des-rivieres-Tshikapa-et-Kasai-Appel-a-laction.pdf>. Accessed 3 Jun 2025.
- [10]. Cui, L., Wang, X., Li, J., Gao, X., Zhang, J., and Liu, Z., 2021. Ecological and health risk assessments and water quality criteria of heavy metals in the Haihe River. *Environ. Pollut.* 290:117971. doi: 10.1016/j.envpol.2021.117971
- [11]. Deko B.O., Omba J.O., Koy R.K., Carvalho, F.P., Poté, J. & Atibu, E.K., 2026. Geochemical assessment of metal and REE concentrations in a region with exploration of petroleum resources, Democratic Republic of the Congo. *Sci Rep.* <https://doi.org/10.1038/s41598-026-62478-3>
- [12]. Förstner, U., Ahlf, W., Calmano, W., 1989. Studies on the transfer of heavy metals between sedimentary phases with a multi-chamber device: combined effects of salinity and redox variation. *Mar Chem.* 28:145–58. [https://doi.org/10.1016/0304-4203\(89\)90192-8](https://doi.org/10.1016/0304-4203(89)90192-8)
- [13]. Groenendyk, D.G., Ferré, T.P.A., Thorp, K.R., Rice, A.K., 2015. Hydrologic-Process-Based Soil Texture Classifications for Improved Visualization of Landscape Function. *PLoS ONE* 10(6): e0131299. doi:10.1371/journal.pone.0131299
- [14]. Håkanson, L., 1980. An ecological risk index for aquatic pollution control: A sedimentological approach. *Water Research*, 14, 975–1001. [https://doi.org/10.1016/0043-1354\(80\)90143-8](https://doi.org/10.1016/0043-1354(80)90143-8)
- [15]. Hao, M., Zuo, Q., Li, J., Shi, S., Li, B., and Zhao, X., 2022. A comprehensive exploration on distribution, risk assessment, and source quantification of heavy metals in the multi-media environment from Shaying River basin, China. *Ecotoxicol. Environ. Saf.* 231:113190. doi: 10.1016/j.ecoenv.2022.113190
- [16]. Hoang, H. G., Chiang, C. F., Lin, C., Wu, C. Y., Lee, C. W., Cheruiyot, N. K., et al., 2021. Human health risk simulation and assessment of heavy metal contamination in a river affected by industrial activities. *Environ. Pollut.* 285:117414. doi: 10.1016/j.envpol.2021.117414
- [17]. Hu, X., Tlili, A., Schirmer, K., Bao, M., & Bürgmann, H., 2025. Metal concentration in freshwater sediments is linked to microbial biodiversity and community composition. *Environment International*. <https://doi.org/10.1016/j.envint.2025.109465>
- [18]. Islam, M., Han, S. S., Ahmed, M. K., & Masunaga, S., 2014. Assessment of trace metal contamination in water and sediment of some rivers in Bangladesh. *Journal of Water and Environment Technology*, 12, 109–121.
- [19]. Jha Sanjay K., Patra Aditi C., Verma Gopal P., Jha Vivekanand and Aswal Dinesh K., 2024. Uranium standards in drinking water: An examination from scientific and socio-economic standpoints of India. *Environmental Science and Pollution Research* (2024) 31:47461–47474 <https://doi.org/10.1007/s11356-024-34352-0>
- [20]. Jokinen, S. A., Jilbert, T., Tiihonen-Filppula, R., & Koho, K., 2020. Terrestrial organic matter input drives sedimentary trace metal sequestration in a human-impacted boreal estuary. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2020.137047>
- [21]. Kipelo, A.M., Jothilakshmi, M., Al Salah, D.M.M., Ngweme, G.M., Atibu, E.K., Mulaji, C., Sivalingam, P., Poté, J.W., 2025. Contamination level and dietary risk assessment of trace elements in *Brassica pkinensis* Lour. from gardening areas under tropical conditions, *Toxicology Reports*, 15, <https://doi.org/10.1016/j.toxrep.2025.102165>.
- [22]. Lesani, S., Zahera, S. S., Hassanzadeh, E., Fuamba, M., & Sharifinejad, A., 2024. Multi-model assessment of climate change impacts on the streamflow

- conditions in the Kasai River Basin, Central Africa. *Hydrology*, 11(12), Article 207. <https://doi.org/10.3390/hydrology11120207>
- [23]. Maanan M, Zourarah B, Carruesco C, Aajjane A, Naud J., 2004. The distribution of heavy metals in the Sidi Moussa lagoon sediments (Atlantic Moroccan Coast). *J Afr Earth Sci.* 39:473–83. <https://doi.org/10.1016/j.jafrearsci.2004.07.017>.
- [24]. Mabidi, D. M., Mutwejile, H. M., Ngweme, G. N. B., Nienie, A. B., Kayembe, J. M., Lusamba, S. N., Atibu, E. K., Carvalho, F. P., Poté, J. & Poté, J., 2024. Heavy Metals Content and Ecotoxicity of Sediments from the Congo River. *American Journal of Environmental Sciences*, 20(1), 64-77. <https://doi.org/10.3844/ajessp.2024.64.77>
- [25]. Manza, H.M., Kiyombo, G.M., Kuna, S.N., Carvalho, F.P., Atibu, E.K., Poté, J., Mata, H.K., 2026. Trace Metal Levels in Sediments and Muscle of Edible Fish Species from the Kasai River, Democratic Republic of the Congo. *Water Air Soil Pollut*, 237:765 <https://doi.org/10.1007/s11270-026-09437-w>
- [26]. McLennan, S. M., 2001. Relationships between the trace element composition of sedimentary rocks and upper continental crust. *Geochemistry, Geophysics, Geosystems*, 2, 2000GC000109. Available online: <https://agupubs.onlinelibrary.wiley.com/doi/epdf/10.1029/2000GC000109>
- [27]. Norvivor, F.A., Azizi, S., Fuku, X., Atibu, E.K., Idris, A.O., Sibali, L., Maaza, M. and Kamika, I., 2024. Ecological and human health risk of heavy metals in Nubui River: a case of rural remote communities. *Front. Water* 6:1397853. doi: 10.3389/frwa.2024.1397853
- [28]. Paithankar, J. G., Saini, S., Dwivedi, S., Sharma, A., and Chowdhuri, D. K., 2021. Heavy metal associated health hazards: an interplay of oxidative stress and signal transduction. *Chemosphere* 262:128350. doi: 10.1016/j.chemosphere.2020.128350
- [29]. Poté J, Haller L, Loizeau J-L, Bravo AG, Sastre V, Wildi W., 2008. Effects of a sewage treatment plant outlet pipe extension on the distribution of contaminants in the sediments of the Bay of Vidy, Lake Geneva, Switzerland. *Bioresour Technol.* 99:7122–31.
- [30]. Rubio, B., Nombela, M. A., & Vilas, F., 2000. Geochemistry of major and trace elements in sediments of the Ría de Vigo (NW Spain): An assessment of metal pollution. *Marine Pollution Bulletin*, 40, 968–980. [https://doi.org/10.1016/S0025-326X\(00\)00039-4](https://doi.org/10.1016/S0025-326X(00)00039-4)
- [31]. Sager, M, Wiche, O., 2024. Rare Earth Elements (REE): Origins, Dispersion, and Environmental Implications—A Comprehensive Review. *Environments*. 11:24. <https://doi.org/10.3390/environments11020024>.
- [32]. Sharma, A.K., Sharma, M., Sharma, S., Malik, D.S., Sharma, M., and Sharma, A.K., 2024. A systematic review on assessment of heavy metals toxicity in freshwater fish species: current scenario and remedial approaches. *J. Geochem. Explor.* 262:107472. doi: 10.1016/j.gexplo.2024.107472
- [33]. Sigue, C., Yewong, N.E., Yiika, L.P., Tiabou, A.F., Simou, A., Liyouck, P.R., Ndema, M.J. L., & Yen, N.V., 2025. Assessment of contamination, sources and health risks of potentially hazardous elements in surface sediments of Dibang, Cameroon. *Discover Environment*, 3, Article 165. <https://doi.org/10.1007/s44274-025-00363-y>
- [34]. Singh, M., Müller, G., & Singh, I.B., 2002. Heavy metals in freshly deposited stream sediments of rivers associated with urbanization of the Ganga Plain, India. *Water, Air, and Soil Pollution*, 141, 35–54.
- [35]. Tarigan, M., Raji, S., Al-Fatesh, H., Czermak, P., & Ebrahimi, M., 2025. The occurrence of micropollutants in the aquatic environment and technologies for their removal. *Processes*, 13(3), Article 843. <https://doi.org/10.3390/pr13030843>
- [36]. Thevenon, F., de Alencastro, L. F., Loizeau, J. L., Adatte, T., Grandjean, D., Wildi, W., & Pote, J., 2013. A high-resolution historical sediment record of nutrients, trace elements and organochlorines (DDT and PCB) deposition in a drinking water reservoir (Lake Bret, Switzerland) points at local and regional pollutant sources. *Chemosphere*, 90, 2444–2452. <https://doi.org/10.1016/j.chemosphere.2012.11.002>
- [37]. Thevenon, F., Graham N. D., Herbez A., Wildi W., Poté J., 2011. Spatio-temporal distribution of organic and inorganic pollutants from Lake Geneva (Switzerland) reveals strong interacting effects of sewage treatment plant and eutrophication on microbial abundance. *Chemosphere*, Volume 84, Issue 5, July 2011, Pages 609-617. DOI : 10.1016/j.chemosphere.2011.03.051
- [38]. Yan, Y., Wan, B., Mansor, M., Wang, X., Zhang, Q., Kappler, A., & Feng, X., 2022. Co-sorption of metal ions and inorganic anions/organic ligands on environmental minerals: A review. *Science of The Total Environment*. <https://doi.org/10.1016/j.scitotenv.2021.149918>
- [39]. Zhu, J., Sun, Z-X., 2024. Estimation of cation exchange capacity for low-activity clay soil fractions using experimental data from South China. *Agronomy*. 14:2671. <https://doi.org/10.3390/agronomy14112671>.
- [40]. Zlobina, A., Farkhutdinov, I., Carvalho, F.P., Wang, N., Korotchenko, T., Baranovskaya, N., et al., 2022. Impact of environmental radiation on the incidence of cancer and birth defects in regions with high natural radioactivity. *Int J Environ Res Public Health*. 19(14):8643. <https://doi.org/10.3390/ijerph19148643>.